

1 Solution — GIAM 7.4

Problem 5

1.1 Problem

The student government at Lagrange High consists of **24** members chosen from amongst the general student body of **210**. Additionally, there is a steering committee of **5** members chosen from amongst those in student government. Use the multiplication rule to determine two different formulas for the total number of possible governance structures.

1.2 Setup

A *governance structure* consists of two pieces of data:

- G : a **24**-element subset of the **210**-student body — the student government.
- S : a **5**-element subset of G — the steering committee.

Equivalently, a governance structure is a *nested pair* $S \subseteq G \subseteq \text{Student Body}$ with $|S| = 5$, $|G| = 24$, $|\text{Student Body}| = 210$.

We want to count the number of such pairs (G, S) — call this number N — using the multiplication rule in two different orders.

1.3 Formula 1 — Outer-Then-Inner

Build the structure in two stages, choosing G first and then S :

- **Stage A.** Choose the **24**-member student government from **210** students: $\binom{210}{24}$ ways.
- **Stage B.** Choose the **5**-member steering committee from those **24**: $\binom{24}{5}$ ways.

By the multiplication rule,

$$N = \binom{210}{24} \cdot \binom{24}{5}.$$

1.4 Formula 2 — Inner-Then-Outer

Build the structure in two stages, choosing S first and then G :

- **Stage A'.** Choose the 5-member steering committee directly from the 210-student body: $\binom{210}{5}$ ways. (Implicitly, these 5 are *also* in student government, since $S \subseteq G$.)
- **Stage B'.** Complete the student government by choosing the remaining $24 - 5 = 19$ members from the remaining $210 - 5 = 205$ students: $\binom{205}{19}$ ways.

By the multiplication rule,

$$N = \binom{210}{5} \cdot \binom{205}{19}.$$

1.5 Verification — Both Formulas Equal

Expand each in factorial form:

$$\binom{210}{24} \cdot \binom{24}{5} = \frac{210!}{24! \cdot 186!} \cdot \frac{24!}{5! \cdot 19!} = \frac{210!}{5! \cdot 19! \cdot 186!}.$$

$$\binom{210}{5} \cdot \binom{205}{19} = \frac{210!}{5! \cdot 205!} \cdot \frac{205!}{19! \cdot 186!} = \frac{210!}{5! \cdot 19! \cdot 186!}.$$

Both expressions collapse to the *same* compact form

$$N = \frac{210!}{5! \cdot 19! \cdot 186!}.$$

The **24!** in Formula 1 cancels between the two factors; the **205!** in Formula 2 likewise cancels. The denominators **5! · 19! · 186!** that survive correspond to *partitioning* the 210 students into three groups — the 5 steering members, the 19 non-steering student-government members, and the 186 non-government students.

1.6 Remark — Why the Two Formulas Must Agree

The agreement of the two formulas is *not* an algebraic accident: both count the *same set* of governance structures. The multiplication rule says nothing about *which* order to use

— only that the count factors. Different orders of building lead to different intermediate expressions, but the total must coincide.

This is the standard *double-counting* technique:

If a set X can be counted in two natural ways, then the two formulas are equal.

Here, X is the set of governance structures, and the two ways are " G then S " versus " S then G ". Equating the two formulas yields the binomial identity

$$\binom{210}{24} \binom{24}{5} = \binom{210}{5} \binom{205}{19}.$$

This is a special case of the general identity

$$\binom{n}{a} \binom{a}{b} = \binom{n}{b} \binom{n-b}{a-b}$$

(here $n = 210, a = 24, b = 5, a - b = 19, n - b = 205$) — the **subset-of-a-subset identity**.

1.7 A Third Perspective — The Multinomial

The common value $N = \frac{210!}{5! \cdot 19! \cdot 186!}$ is the *trinomial coefficient*

$$N = \binom{210}{5, 19, 186},$$

counting the number of ways to partition **210** students into three *labeled* groups of sizes **5**, **19**, and **186**:

Group	Size
Steering committee	5
Student government but not steering	19
Not in student government	186

Table 1.1

This is a single application of the multinomial coefficient, not a sequence of two binomial choices — yet it counts the same governance structures. The three formulas — Formula 1, Formula 2, and the multinomial — all evaluate to the same integer:

$$\binom{210}{24} \binom{24}{5} = \binom{210}{5} \binom{205}{19} = \binom{210}{5, 19, 186}.$$

1.8 Remark — A Sanity Check on Sizes

The smaller factor $\binom{24}{5} = 42,504$ is easy to verify:

$$\binom{24}{5} = \frac{24 \cdot 23 \cdot 22 \cdot 21 \cdot 20}{5!} = \frac{5,100,480}{120} = 42,504.$$

And $\binom{210}{5} = \frac{210 \cdot 209 \cdot 208 \cdot 207 \cdot 206}{120} = 3,244,032,787 \approx 3.24 \times 10^9$.

The total N itself is *astronomically* large — $\binom{210}{24}$ alone is approximately 10^{30} , and multiplied by $\binom{24}{5} \approx 4 \times 10^4$ gives $N \approx 10^{34}$ or so. The point of the exercise is *not* to compute N numerically but to demonstrate that *two natural counting orders give the same answer* — yielding a non-trivial binomial identity along the way.

1.9 Remark — The General Counting Principle

The “build in stages” technique used in both formulas is the multiplication rule applied to *sequential* choices, where the number of options at stage k may depend on earlier choices but not on *which* earlier choices were made. Here:

- In Formula 1, the choice at stage B (steering committee) has $\binom{24}{5}$ options regardless of *which* 24 students were chosen in stage A.
- In Formula 2, the choice at stage B' (rest of government) has $\binom{205}{19}$ options regardless of *which* 5 students were chosen in stage A'.

Whenever a problem admits two such “build orders,” the resulting formulas must be equal, and the equality is often a useful identity in its own right.